400

No. of routes India expects to cover under its high-speed rail network

₹79,000

CRORE

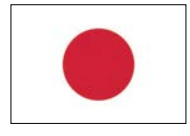
The soft loan being provided by Japan for India's Mumbai-Ahmedabad bullet train project



AI generated image by BANDEEP SINGH

in each direction initially, which will be subsequently increased to 92,900 passengers per day in each direction by 2053

- In the future, India aims to make high-speed trains domestically
- The initial plan is use Indian Railways' own manufacturing capacities in collaboration with Japan



JAPAN

- Japan will fund 80 per cent of the cost of the Mumbai–Ahmedabad HSR project through a soft loan of Rs 79,000 crore
- In May this year, 20 experts from the Japan Railway Technical Services (JRTS) trained 1,000 Indian engineers for the Mumbai–Ahmed HSR project
- Japan will also provide 18 of its Shinkansen-style train coaches to India for the project

Delhi-Chandigarh-Amritsar. The plan is to purchase 18 Shinkansen trains from Japan and gradually upgrade Indian factories and the industry ecosystem to indigenously produce them.

A high-speed rail network will undoubtedly aid the economy too. According to a study conducted by the London School of Economics and Political Science and the University of Hamburg in 2008, cities that are connected to high-speed rails tend to see a rise in their GDP by at least 2.7 per cent compared to their neighbours.

WHAT INDIA NEEDS TO MASTER

HSR networks require special technology. In May, around 1,000-odd engineers were trained to operate the ballast-less Slab Track system (popularly known as J-Slab track system) that is used for Shinkansens. HSRs use a higher number of motored axles too, which permits higher acceleration so that the train does not lose much time at stops. Notably, Shinkansen lines have more stops in proportion to their length than high-speed lines elsewhere in Europe and China. Also, these high-speed trains will require a dedicated standard gauge line, which cannot be used by the existing trains in India that are broad-gauged, thus reducing interoperability—unless India decides in the future to shift to variable gauges. The challenge is to meet these and many more such technical requirements in a time-bound manner.

India will also have to tackle many other infrastructural challenges proactively, while also optimising cost and prioritising safety. For example, India's current semi-high-speed trains often can't run at their peak speed due to the condition of the tracks. Other bottlenecks such as outdated signalling systems and train delays will have to be resolved too to make India's high-speed rail project a truly world-class venture. ■